

GROUP ASSIGNMENT REPORT

AI-Based Smart Healthcare Diagnosis and Disease Prediction System

Saveetha School of Engineering

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Academic Year / Batch: 2026-2029

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SDG Mapping	SDG 3 – Good Health and Well-Being
Industry / Societal Relevance	AI-Based Smart Healthcare Diagnosis and Disease Prediction System

Intelligent Knowledge-Based Navigation for an Autonomous Mobile Robot (AMR)

CO4 / BL4 – Analytical Answer

Introduction

An Autonomous Mobile Robot (AMR) in a modern manufacturing plant continuously receives data from cameras, LiDAR, ultrasonic sensors, proximity sensors, wheel encoders, and other sources. These sensors provide observations such as the presence of an obstacle, the width of a passage, the movement of an object, or the availability of a clear corridor.

However, raw sensor observations alone do not constitute intelligent behavior. The robot must interpret these observations, combine them with previous knowledge and operational rules, reason about the current situation, and select a safe and efficient action.

Example: LiDAR detects an object → object is interpreted as a pallet → pallet blocks the planned route → alternative route is required → robot replans its path → robot changes direction.

Thus, intelligent navigation can be represented as:

Observation → Information → Knowledge → Reasoning → Decision → Action → Knowledge Update

1. Observation and Knowledge Representation

1.1 Raw observation, interpreted information, and knowledge

Level	Meaning	AMR example
Raw observation	Direct sensor reading	LiDAR detects points at 1.5 m
Interpreted information	Meaning assigned to sensor data	An object is detected 1.5 m ahead
Knowledge	Generalized understanding or rule used for reasoning	If an object blocks the planned path, stop or find an alternative route

Raw observation

Raw observations are numerical or sensory measurements obtained directly from sensors.

- Camera detects a visual object.
- LiDAR returns a cluster of points.

- Ultrasonic sensor measures a short distance.
- Proximity sensor becomes active.
- Wheel encoder reports movement.

A single reading does not necessarily tell the robot what the situation means.

Interpreted information

The robot processes sensor data to identify meaningful environmental conditions. For example, LiDAR detects a cluster of points 1.2 m ahead → there is probably an object in the robot's planned path. Similarly, a camera image can be processed as object recognition → worker detected.

Useful knowledge

Knowledge connects environmental conditions with consequences and possible actions. For example: If an obstacle occupies the planned navigation corridor, the current route is blocked and the robot must stop, avoid the obstacle, or select an alternative route.

Knowledge can be represented using IF–THEN rules, semantic maps, occupancy grids, object and environment models, knowledge graphs, state representations, probabilistic models, and learned policies.

1.2 Representing observations as knowledge

Obstacle detected: Instead of simply storing 'obstacle = true', the robot can represent the object using properties such as position, size, confidence, and velocity. This allows the system to reason about what the object is, where it is, how large it is, how confidently it was detected, and whether it is moving.

Narrow path detected: The robot can represent path width and compare it with the required safe operating width. If path width is below the safe width, it reduces speed and increases obstacle-clearance constraints.

Open path detected: The robot can establish that the path is sufficiently wide and free from detected hazards. If the path is clear and confidence is high, it may increase speed within permitted limits.

1.3 Industrial situations

A. Pallet blocking the normal route

Observation: object detected → Interpretation: pallet occupies route → Knowledge: planned path is blocked → Reasoning: determine whether the pallet is temporary and whether another safe route exists → Decision: stop/replan → Action: use an alternative corridor.

B. Narrow passage between two machines

LiDAR detects two nearby machine boundaries and calculates a small available width. The robot can infer that the passage is traversable only with reduced speed and increased positional accuracy. If width is below the safe clearance requirement, the passage is classified as non-traversable.

C. Temporarily restricted production area

The robot combines map information, access-control information, sensor observations, and temporary digital restrictions. Knowledge becomes: Zone 4 is restricted. Decision: do not enter Zone 4 and select another route.

D. Clear and wide transportation corridor

Sensors consistently indicate sufficient width, no obstacles, no humans, no forklifts, and acceptable conditions. The corridor is suitable for normal-speed transportation, subject to plant safety limits.

E. Moving workers or forklifts

If a detected object changes position over time, the robot can infer that it is moving. If its predicted trajectory intersects the robot's path, the robot should reduce speed, stop, or wait until the path is safe.

2. Knowledge-Based Decision Making

The robot can maintain a knowledge base containing facts, rules, maps, safety constraints, and learned information.

IF obstacle blocks route THEN stop or select alternative route.

IF path width is below normal operating width THEN reduce speed.

IF path is wide AND clear AND confidence is high THEN allow higher permitted speed.

IF human is detected within a safety zone THEN slow down or stop.

IF a moving object approaches the planned trajectory THEN predict collision risk and modify the trajectory.

IF environmental uncertainty is high THEN select a conservative action.

IF all nearby routes are unsafe THEN stop and request assistance.

These rules should not operate independently.

Combining multiple observations

- LiDAR: object detected.
- Camera: object appears to be a forklift.
- Tracking system: object is moving.
- Map: forklift crossing zone.
- Robot trajectory: collision predicted.

Together, these observations produce stronger knowledge: a moving forklift is crossing the robot's planned path and presents a potential collision risk. The resulting decision may be: reduce speed → stop before the intersection → wait → verify path → continue. This illustrates why sensor fusion and contextual reasoning are essential.

3. Transformation of Observations into Knowledge

A robot should not treat every observation as an isolated event. Repeated observations can help establish more general knowledge.

Observation → Interpretation → Knowledge → Decision

Temporary versus permanent obstacle

If a pallet appears on a route and later disappears, the robot can infer that it was temporary. If a machine structure remains fixed at the same location over repeated observations, it is likely a permanent environmental feature.

Genuine narrow path versus false detection

If one camera image makes a corridor appear narrow but LiDAR and repeated measurements show adequate width, the initial classification was probably a visual error. If multiple sensors repeatedly measure insufficient width, confidence in the narrow-path interpretation increases.

Open versus partially obstructed path

If the camera sees a clear corridor but LiDAR detects an object near one side, the robot should represent the state as partially obstructed rather than completely open.

Stationary versus moving object

Repeated position measurements allow velocity estimation. A nearly unchanged position suggests a stationary object, while consistent changes in position suggest a moving object. The robot can then predict where the object will be when the robot reaches that location.

Safe versus uncertain environment

Safety depends on sensor coverage, sensor reliability, detection confidence, map accuracy, communication status, visibility, and dynamic-object predictions. If these are unreliable, the robot should represent the environment as uncertain rather than assuming it is safe.

4. Handling Noisy Observations

Industrial environments are challenging because sensors can be affected by dust, smoke, poor lighting, reflective surfaces, vibration, machinery, metal surfaces, occlusion, electromagnetic interference, sensor aging, and communication failures.

Therefore, sensor observations should normally have an associated confidence or uncertainty value. For example, 'obstacle detected, confidence 0.92' provides stronger evidence than 'obstacle detected, confidence 0.35'.

False obstacle detection

A reflective surface may cause LiDAR to produce an apparent obstacle. If the robot immediately changes direction, it may waste time or disrupt production. The robot should compare the reading with other sensors and previous observations. If the evidence repeatedly disappears, confidence in the obstacle hypothesis can decrease. Because navigation is safety-critical, uncertainty should still lead to cautious verification.

Missed obstacle

This is more dangerous. An actual pallet may be missed because of sensor occlusion, dust, blind spots, or communication failure. The robot may incorrectly believe the path is open. Multiple sensors, overlapping sensing regions, obstacle prediction, safety margins, and emergency stopping mechanisms are therefore important.

Incorrect open-path classification

If a narrow or partially blocked path is classified as open, the robot could travel too quickly and collide with an obstacle. High-speed operation should require sufficiently high confidence in the path state.

Conflicting sensor readings

For example, the camera says clear while LiDAR and ultrasonic sensors detect an object. The robot should use sensor fusion, sensor reliability, temporal consistency, and safety rules rather than simply trusting one reading.

5. Handling Incomplete Knowledge

A robot can never assume that it has complete knowledge of the factory.

Path outside sensor range

The robot may see that the next 5 m are clear but have no information about what lies beyond that region. It should recognize that the future path is partially unknown and use sensing and braking distances to maintain safety.

Forklift suddenly enters the route

The robot must continuously update its environment model rather than relying entirely on a previously planned route. It should detect, track, predict, slow or stop, replan, and continue when safe.

Worker hidden behind a machine

The robot may not directly observe a worker because of occlusion. Contextual knowledge can indicate that people may emerge from the area. The robot can reduce speed, increase clearance, or stop before an occluded intersection.

Sensor temporarily unavailable

If LiDAR becomes unavailable, the robot should not behave as if nothing has changed. It can use redundant sensors, reduce speed, restrict operation, or enter a safe state.

Insufficient information

Unknown should not automatically be treated as safe. If the robot cannot confidently establish that a route is safe, it should select a safer action, gather more information, reduce speed, or stop if necessary.

6. Conflicting and Ambiguous Observations

Conflicting observations occur when sensors provide different interpretations. For example, a camera may identify a wide clear corridor while LiDAR detects an object on the right side.

Possible explanations include a camera miss, LiDAR reflection, partial occlusion, or incorrect segmentation. The robot should maintain uncertainty until enough evidence is available.

Ambiguous object classification

An object may be a stationary machine component, pallet, worker, or forklift. The consequences are different. A moving worker may require immediate stopping, whereas a stationary pallet may only require rerouting. When classification confidence is low, the robot should use a conservative strategy.

A useful principle is: Object detected + classification uncertain + collision possibility = treat as a potential hazard.

Handling conflicting information

- Sensor reliability
- Confidence scores
- Temporal consistency
- Spatial consistency
- Historical observations
- Environmental context
- Safety rules

Sensor fusion combines these factors to estimate the most reliable environmental state.

7. Safety and Industrial Constraints

Safety should have higher priority than productivity because a collision can cause worker injury, equipment damage, material loss, production interruption, robot damage, and safety-system violations.

A practical priority hierarchy is: Human safety → collision avoidance → equipment protection → operational constraints → productivity optimization.

Sudden obstacle

Detect → safe braking/stop → assess → replan. The robot should not prioritize its delivery schedule over collision avoidance.

Human enters operating area

The robot should apply the configured safety response, such as slowing down, stopping, or maintaining safe separation, and resume only when the area is confirmed safe.

Path safety cannot be determined

Stop or move into a predefined safe state rather than guessing.

Multiple routes

Evaluate collision risk, route length, congestion, path width, restricted areas, moving-object predictions, surface conditions, energy use, and delivery priority. Select the safest feasible route first, then optimize efficiency.

Speed versus productivity

Higher speed reduces transportation time but also reduces reaction time and increases stopping distance. Speed should therefore depend on environmental conditions and remain within certified plant limits.

Environment	Appropriate behavior
Wide, clear, high-confidence corridor	Normal/higher permitted speed
Narrow corridor	Reduced speed
Moving workers nearby	Very low speed/stop as required
Uncertain environment	Conservative speed
Obstacle on route	Stop/replan
Human in protective zone	Safe stop/controlled response

8. Knowledge Updating

The factory environment is dynamic, so the robot's knowledge cannot remain static.

Confirm existing knowledge

Repeatedly observing a fixed machine at the same position confirms a permanent map feature.

Correct previous knowledge

If the map says a corridor is open but repeated observations show that it is blocked, the map should be updated.

Add new knowledge

A newly installed machine can become a new environmental feature.

Remove outdated knowledge

A temporary construction barrier can eventually disappear and its restriction can be removed.

Change the current decision

A robot may select Route A, but if a forklift enters Route A, new knowledge makes it unsafe and the robot can switch to Route B.

Continuous updating is essential because an industrial plant is a dynamic environment rather than a fixed map.

9. Overall Analysis

The complete intelligent navigation process is:

Sensors → Raw Observations → Data Processing / Fusion → Interpretation → Environmental Knowledge → Uncertainty Estimation → Reasoning → Safety + Operational Evaluation → Decision → Action → Knowledge Update

Example: Forklift crossing the robot's route

1. Sensor observation: Camera and LiDAR detect an object.
2. Interpretation: The object is identified as a moving forklift.
3. Knowledge formation: The forklift trajectory intersects the robot's planned path.
4. Reasoning: Collision risk is high.
5. Decision: Slow down or stop and wait.
6. Action: Robot stops before the intersection.
7. New observation: Forklift has passed.
8. Knowledge update: Route is clear with high confidence.
9. Decision: Resume navigation.

Handling the main types of uncertainty

Problem	Example	Appropriate response
Noise	LiDAR gives inconsistent distances	Filter/fuse observations
False positive	Obstacle reported when none exists	Verify using other sensors/time
False negative	Real pallet missed	Use redundancy and safety margins
Incomplete information	Path beyond sensing range unknown	Reduce risk/speed or obtain more information
Ambiguity	Object could be worker or machine part	Use conservative interpretation
Conflict	Camera says clear, LiDAR says blocked	Fuse evidence and estimate confidence

Dynamic change	Forklift enters route	Update world model immediately
Sensor failure	LiDAR unavailable	Use redundancy or enter safe state

Why knowledge-based navigation is better than simple reactive navigation

A purely reactive robot might implement 'Obstacle → turn'. This is inadequate for a modern manufacturing plant.

An intelligent knowledge-based robot can reason that a detected object is probably a temporary pallet, the current route is blocked, another route is available but narrow, a worker is nearby, and a longer route has lower uncertainty. It can therefore choose the safest feasible route rather than simply reacting to one sensor reading.

This represents the transition from simple reaction to intelligent decision-making.

Conclusion

An intelligent AMR must transform raw sensor observations into structured, uncertain, and continuously updated knowledge. The complete process is: Sensor Observation → Interpretation → Knowledge Formation → Reasoning → Decision → Action → Knowledge Update.

Raw observations such as obstacle detection, narrow-path detection, and open-path detection become useful only after they are interpreted in relation to the robot's position, map, previous observations, object behavior, safety constraints, and operational objectives.

Knowledge representation allows the robot to understand permanent versus temporary obstacles, narrow versus sufficiently wide paths, stationary versus moving objects, safe versus uncertain environments, and restricted versus accessible areas.

Because manufacturing environments contain noise, incomplete information, sensor conflicts, ambiguous objects, and constantly changing conditions, the robot should not treat every observation as certain. Sensor fusion, confidence estimation, temporal reasoning, redundancy, conservative decision-making, and continuous knowledge updating are therefore essential.

Most importantly, safety must be treated as a hard constraint rather than merely another optimization objective. Productivity can be optimized only after safety requirements have been satisfied.

Consequently, the knowledge-based approach enables the AMR to achieve safer navigation, better collision avoidance, efficient route selection, adaptation to dynamic factory conditions, robust decisions under uncertainty, continuous knowledge updating, and greater autonomy in smart manufacturing environments.

In this way, the AMR evolves from a machine that merely reacts to sensor readings into an intelligent system that understands its environment, reasons about uncertainty, updates its knowledge, and takes safe and purposeful actions.

One-Page Individual Reflection

Working on the Intelligent Autonomous Mobile Robot (AMR) assignment helped me understand how an autonomous system can transform raw and uncertain sensor observations into meaningful knowledge and safe decisions. One of the major design decisions was to use a **knowledge-based, multi-sensor reasoning approach** rather than depending on a single sensor or a purely reactive response. The proposed approach combines observations from sensors such as LiDAR, cameras, and proximity sensors, interprets them using confidence and uncertainty, updates the robot's knowledge base, and then applies decision rules to select an appropriate action. This design was particularly important for handling situations such as obstacles, moving workers or forklifts, blocked paths, narrow passages, and temporarily restricted areas.

A major challenge was dealing with **noisy, incomplete, conflicting, and ambiguous observations**. In a real manufacturing environment, sensors may produce inaccurate measurements, temporarily lose information, or provide different interpretations of the same situation. I learned that an autonomous robot should not treat every observation as completely reliable. Instead, observations need to be evaluated using confidence levels, sensor fusion, contextual information, and safety priorities. Another important challenge was designing decisions that remain safe even when the robot does not have complete knowledge of its environment. This led to the understanding that when uncertainty is high, the robot should choose a conservative action such as slowing down, stopping, waiting for additional information, or selecting an alternative route.

The assignment strengthened my understanding of several course concepts, particularly **knowledge representation, reasoning, uncertainty handling, decision making, rule-based systems, and knowledge updating**. I learned how an observation such as “an obstacle is detected ahead” can be transformed into knowledge such as “the current path is blocked,” which can then support a decision such as “stop and re-plan.” The assignment also showed me the importance of continuously updating the knowledge base because the environment of an industrial plant is dynamic.

The work has a clear connection with **Sustainable Development Goal (SDG) 9 – Industry, Innovation and Infrastructure**, as autonomous mobile robots can support smarter, safer, and more efficient industrial operations. It also relates to **SDG 8 – Decent Work and Economic Growth** by demonstrating how intelligent automation can assist workers, improve material movement, and reduce exposure to hazardous situations. The emphasis on safety, reliability, and responsible automation helped me understand that technological development must consider both performance and its impact on people and the workplace.

Overall, this assignment contributed to attaining the mapped **Course Outcome (CO)** by requiring me to apply theoretical concepts to a practical engineering problem, analyze requirements and constraints, develop a systematic solution, and justify design decisions. It improved my ability to reason about uncertain real-world environments and demonstrated how concepts learned in the course can be integrated into an intelligent autonomous system. The experience also improved my problem-solving, analytical thinking, and technical documentation skills.